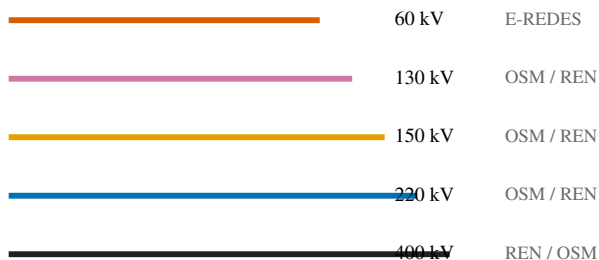


(a) Voltage-specific source layers

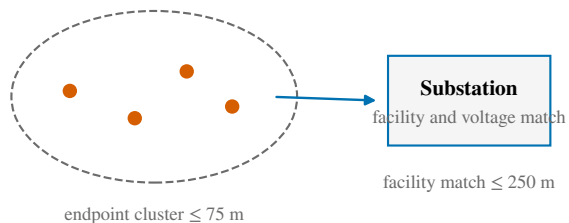


(c) Explicit nodes and parallel circuits



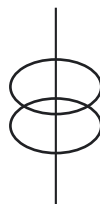
parallel paths remain distinct; junctions remain explicit

(b) Endpoint clustering and facility matching

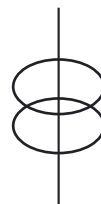


(d) Transformer connection rules

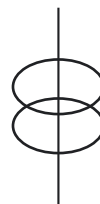
voltage pair + facility evidence + distance ≤ 1 km



220/60 kV



220/150 kV



400/220 kV